Abderrahmane Er-raqabi

AbderrahmaneErraqabi | Abderrahmane Er-Raqabi | abderrahmaneer-raqabi.com |
abderrahmane.erraqabi@gmail.com | 514-806-3430

About

Electrical Engineering student at Polytechnique Montréal focused on embedded systems, electronics, and hardware-software integration. Experience in instrumentation, signal processing, control, and electronic prototyping. Member of the Esteban team and VP External Relations at GDG Polytechnique Montréal.

Education

2025 – Present **Bachelor of Electrical Engineering** — Polytechnique Montréal (Montreal, QC)
Main interests: embedded systems, electronic design, signal processing, and hardware-software integration. Member of the Esteban team and VP External Relations at GDG Polytechnique Montréal.

2023 – 2025 **Diploma of College Studies (DEC) in Natural Sciences** — Collège Bois-de-Boulogne (Montreal, QC)
Scientific background focused on physics, chemistry, mathematics, and programming.

Technical Skills

Languages	C/C++, Python, MATLAB, Rust
Embedded Systems	ESP32/ESP32-S3, STM32, Arduino, Raspberry Pi, FreeRTOS
Interfaces	SPI, I2C, UART, CAN, Wi-Fi, BLE, NFC/RFID, LoRa/Sub-GHz, PWM, ADC
Tools	LTspice, MATLAB/Simulink, KiCad, AutoCAD Electrical, RapidHarness, Git/GitHub
Core Skills	Circuit and PCB design, instrumentation, analog filtering, wiring, embedded systems integration and validation
Interests	Embedded systems, power electronics, control, RF, signal processing, and embedded AI

Projects

ESP32-S3Zero, Multi-Interface Embedded Platform — ESP32-S3, NFC, FreeRTOS, RF, Wi-Fi/BLE *2026 – Ongoing*

- Designed an ESP32-S3 platform integrating Wi-Fi/BLE, NFC, LoRa/Sub-GHz, IR, microSD, and a TFT display.
- Developed the firmware architecture and electrical schematic with power management; hardware build in progress.
- Modularized firmware for communications, peripherals, UI, and power management.

Frequency-Multiplexed Analog System — ELE2003 — LTspice, Analog Electronics, Signal Processing *2026*

- Designed an analog system multiplexing two audio signals and a control signal on one line.
- Designed and tuned filters and a modulation/demodulation chain in LTspice, then validated them in the lab.

ESP32 AI Camera Web Server — ESP32, Edge Impulse, Embedded Vision *2026*

- Developed an embedded vision system running a quantized int8 Edge Impulse model on an ESP32.
- Integrated image acquisition, real-time inference, and video streaming through a local web server.
- GitHub — ESP32 AI Camera

AM Radio Receiver — RF, Analog Electronics, LC, LM386 *2026*

- Designed a 540–1600 kHz AM receiver using a loop antenna and an LC resonant tuning circuit.
- Integrated 1N34A envelope detection and LM386 amplification, then validated the receiver.
- GitHub — AM Radio Receiver

MedBot — Automated Medication Dispenser — StarHack 2026 — Arduino, C++, Embedded Electronics *2026*

- Designed an autonomous embedded system for medication reminders and automated pill dispensing.
- Integrated RTC, LCD, keypad, servo, and a motor into a complete hardware-software architecture.
- GitHub — MedBot / StarHack 2026

ELE1001 — BCD Keypad Interface — Arduino, Digital Logic, BCD *2025*

- Developed a 4×4 keypad interface generating BCD outputs and synchronization signals for external logic modules.
- GitHub — ELE1001

Work Experience

Electronics / PCB Team — Project Esteban, Polytechnique Montréal Aug 2025 – Present
Design and integration of embedded electronics for a solar car, including schematics, PCBs, power systems, wiring, and hardware validation.

VP External Relations — Google Developer Group Jan 2026 – Present
Industry partnerships, sponsorship coordination, and technical events connecting students with professionals.

Private Tutor — Mathematics and Physics Aug 2025 – Present
Tutoring from secondary school to university, focused on problem solving and conceptual understanding.

Scientific Intern — Water Quality Screening — City of Montreal May 2026 – Aug 2026
Over 300 field analyses: sampling, Kemio electrochemical measurements, result interpretation, and data compilation.

Sales Manager — SoftMoc Jun 2023 – Present
Team supervision, performance monitoring, and operations management in a high-volume environment.

Certifications

Battery Systems Safety	Institut du véhicule innovant (IVI), Saint-Jérôme — *2026*
Quantum Programming — Grover's Algorithm	Institut quantique, Université de Sherbrooke — *2025*
Cloud Computing Security	Canadian Mobility and Aerospace Institute (CMAI/IMACA) — *2025*
WHMIS 2015	Polytechnique Montréal / LeCampus — *2025*
Respiratory Protection	Polytechnique Montréal — *2025*